

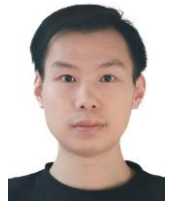
Xiaoyong (William) Ni

PhD Candidate, EPFL Neuro-X Institute | ML for Neural Decoding & Brain-Computer Interfaces

(+41) 76 539 53 84 | xiaoyong.ni@epfl.ch | Geneva, Switzerland

[Google Scholar](#) (1,100+ citations) | [LinkedIn](#) | [GitHub: KalmanNet](#) (450+ stars)

ML researcher working on **model-based deep learning, unsupervised adaptation, and time-series decoding** for brain signals. Co-developer of **KalmanNet** (870+ citations, IEEE SPS Best Paper nomination); 9 papers incl. NeurIPS, IEEE TSP, ICASSP and 2 EMBC orals, plus 2 first-author IEEE TBME papers under review. Seeking a **Summer 2027 research internship**.



EDUCATION

EPFL, Switzerland Jun 2024 – Present
PhD, Neuro-X Institute (Integrated Neurotechnologies Lab) — ML for brain signal processing and decoding

ETH Zurich, Switzerland Sep 2020 – Jun 2023
MSc in Electrical Engineering and Information Technology | GPA: 5.5/6.0 | Thesis: 5.75/6.0 (Excellent)

University of Glasgow, UK & UESTC, Chengdu, China Sep 2016 – Jul 2020
BEng in Communication Engineering (Dual Degree, Electronics & Electrical Eng.) | GPA: 3.96/4.0 (Rank 1/126) | First Class Honours

RESEARCH EXPERIENCE

Integrated Neurotechnologies Lab, EPFL — PhD Candidate Jun 2024 – Present

- **Real-time stimulation artifact cancellation (ReCAST; first author, under review at IEEE TBME):** showed that stimulation artifacts lie in a stable, low-dimensional spatial subspace (≤ 3 components hold $\geq 99.8\%$ of artifact energy) separable from neural activity; built a recursive subspace-tracking algorithm with adaptive dimension selection that cancels artifacts **sample by sample, with no look-ahead**. Validated on real ECoG/DBS recordings, synthetic benchmarks and saline bench experiments; seizure-detection F1 after cleaning **matched clean data (97.4% vs. 97.3%)** with no decoder retraining, the best of all compared methods.
- **Reliable metrics for neural signal denoising (FND; first author, under review at IEEE TBME):** built a meta-evaluation framework that tests FID-style metrics for EEG/iEEG on monotonicity, sampling stability and robustness to data scarcity, across embeddings ranging from hand-crafted features to pre-trained **neural foundation models** (BrainBERT, Brant, BIOT, etc.); showed that raw-signal Fréchet distance is the most reliable (**up to 100% rank agreement** with ground truth) and that using it as an unsupervised training loss improves denoising.
- **Supervised MSc student — PULSE (EMBC 2026 oral, corresponding author):** unsupervised deep learning that combines physics constraints with a pre-trained neural “expert” prior to remove stimulation artifacts from iEEG without ground truth; **>23 dB artifact suppression, ≈ 13 dB SNR gain**.
- **Supervised MSc student — iNeuroTune (EMBC 2026 oral, corresponding author):** interpretable neural-augmented Kalman filter decoder that adapts to neural drift **without new labels** via reliable pseudolabel selection; **R^2 0.81 $\rightarrow$ 0.83** on Neural Latents Benchmark MC_Maze using only 25% labeled data, while an LLM self-training baseline (AcTune) degraded to 0.80.
- Building decoders from intracranial neural signals to movement for BCIs that restore function in paralyzed patients, toward closed-loop stimulation systems.

Signal and Information Processing Lab, ETH Zurich — Research Assistant & MSc Thesis Nov 2021 – May 2024

- **KalmanNet:** co-developed a hybrid RNN-aided Kalman filter that learns the Kalman gain from data under partially known, nonlinear dynamics; built the simulation and benchmarking framework. IEEE TSP 2022, **Best Paper Award nomination, 870+ citations**, open-source repo with **450+ GitHub stars**.
- **RTSNet:** extended model-based deep learning from filtering to smoothing (learned Rauch–Tung–Striebel smoother); first author at ICASSP 2022, journal version in IEEE TSP 2023.
- **Adaptive KalmanNet (MSc thesis):** introduced lightweight fine-tuning inspired by LLM adaptation, letting the learned filter adapt quickly to time-varying noise statistics while keeping its model-based structure; first author, ICASSP 2024 (80+ citations).
- Jan–May 2024: contributed to a NeurIPS 2024 study comparing deep-learning and explainable decoders for brain-machine interfaces, while applying to PhD programs.

Earlier research — Wireless communications 2019

- **Penn State University** (Jul–Oct 2019): Age-of-Information analysis and MDP-based scheduling for energy-harvesting cognitive radio networks (IEEE MASS 2019).
- **UESTC** (Feb–Jun 2019): asymmetric PSK/QAM modulation design for simultaneous wireless information and power transfer.

SKILLS

ML: PyTorch; self-supervised / unsupervised learning; model-based deep learning; fine-tuning and adaptation

Signal processing: Kalman filtering / smoothing; state-space models; artifact removal; time-series decoding

Neural data: intracranial recordings (ECoG / LFP / spikes / DBS) and EEG; real-time decoding; closed-loop stimulation
Programming & tools: Python, MATLAB, C; Git, Docker
Languages: Chinese (native), English (fluent)

PUBLICATIONS

*Bold = me; * = corresponding author. Full list on [Google Scholar](#).*

- **Under review, IEEE TBME** — *Real-time Stimulation Artifact Cancellation via Differential Artifact Subspace Tracking*. **X. Ni**, O. Etesami, D. Alex, M. Shoaran.
- **Under review, IEEE TBME** — *FND: A Meta-Evaluation Framework for Reliable Assessment of Neural Signal Denoising*. **X. Ni**, T. Zhu, M. Shoaran.
- **EMBC 2026, Oral** — *Physics Constraints and Expert Regularization for Unsupervised Stimulation Artifact Removal*. T. H. Bui, **X. Ni***, M. Shoaran. [\[code\]](#)
- **EMBC 2026, Oral** — *iNeuroTune: Interpretable Neural Data Selection for Unsupervised Decoder Fine-Tuning*. A. Wisniewska, **X. Ni***, M. Shoaran.
- **NeurIPS 2024** — *Exploring the Trade-off Between Deep-Learning and Explainable Models for Brain-Machine Interfaces*. L. H. Cubillos, G. Revach, M. J. Mender, et al. (incl. **X. Ni**).
- **ICASSP 2024** — *Adaptive KalmanNet: Data-Driven Kalman Filter with Fast Adaptation*. **X. Ni**, G. Revach, N. Shlezinger. [\[code\]](#)
- **IEEE TSP 2023** — *RTSNet: Learning to Smooth in Partially Known State-Space Models*. G. Revach, **X. Ni**, N. Shlezinger, R. J. G. van Sloun, Y. C. Eldar. [\[code\]](#)
- **ICASSP 2022** — *RTSNet: Deep Learning Aided Kalman Smoothing*. **X. Ni**, G. Revach, N. Shlezinger, R. J. G. van Sloun, Y. C. Eldar. [\[code\]](#)
- **IEEE TSP 2022** — *KalmanNet: Neural Network Aided Kalman Filtering for Partially Known Dynamics*. G. Revach, N. Shlezinger, **X. Ni**, A. L. Escoriza, R. J. G. van Sloun, Y. C. Eldar. **IEEE SPS Best Paper Award nomination; 870+ citations.** [\[code\]](#)
- **EUSIPCO 2022** — *Unsupervised Learned Kalman Filtering*. G. Revach, N. Shlezinger, T. Locher, **X. Ni**, R. J. G. van Sloun, Y. C. Eldar.
- **IEEE MASS 2019** — *Age of Information for Wireless Energy Harvesting Secondary Users in Cognitive Radio Networks*. S. Leng, **X. Ni**, A. Yener.

HONORS & SERVICE

ICML Gold Reviewer	2026
IEEE Signal Processing Society Best Paper Award Nomination (KalmanNet)	2022
Outstanding Graduate of Sichuan Province	2020
Department Outstanding Academic Scholarship, 3 consecutive years (Top 1%)	2017 – 2019
First Prize, UESTC “Ecovacs” Robot Competition (1st of 143)	2018